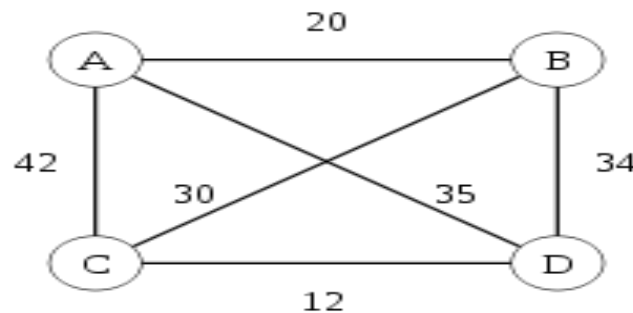


Travelling Salesman Problem

- Given a list of cities and the distances between each pair of cities.
- What is the shortest possible routes that visits each city exactly once and returns to the original city?
- Exhaustive search is not feasible.
- It is an NP-hard problem in combinatorial optimization.



Techniques

- Brute-force approach
 - ❑ Check every possible route and return the shortest route.
 - ❑ Time complexity – $O(n!)$
 - ❑ $10!$ is 3,628,800
 - ❑ $20!$ is 2432902008176640000
 - ❑ $30!$ is ??

Techniques

■ Dynamic Programming

- ❑ Keep optimal sub-routes of certain length with their end-point.
- ❑ Then add a point to the routes and figure out optimal sub-routes that are now one point longer than in the previous step.
- ❑ Keep doing this until we get to the full route.
- ❑ Time complexity $-O(n^2 \cdot 2^n)$
- ❑ $10^2 2^{10} = 102,400$
- ❑ $20^2 2^{20} = 419430400$
- ❑ $30^2 2^{30} = 966367641600$
- ❑ $40^2 2^{40} = ??$

Heuristic Approach

- A heuristic is a technique designed for solving a problem **more quickly** when classic methods are too slow
- Or, for finding an approximate solution when **classic methods fail to find any exact solution.**
- Results about NP-hardness in theoretical computer science make **heuristics the only viable option for a variety of complex optimization problems** that need to be routinely solved in real-world applications.

Heuristic Approach (cont.)

- The trade-off criteria for deciding **whether to use a heuristic for solving** a given problem include the following:
- **Optimality:** When several solutions exist for a given problem, does the heuristic guarantee that the best solution will be found? Is it actually necessary to find the best solution?
- **Completeness:** When several solutions exist for a given problem, can the heuristic find them all? Do we actually need all solutions? Many heuristics are only meant to find one solution.

Heuristic Approach (cont.)

- ***Accuracy and precision:*** Can the heuristic provide a confidence interval for the purported solution? Is the error bar on the solution unreasonably large?
- ***Execution time:*** Is this the best-known heuristic for solving this type of problem? Some heuristics converge faster than others. Some heuristics are only marginally quicker than classic methods.

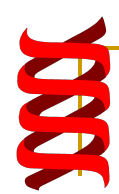
- **Bio-inspired heuristic approach:**
 - ❑ Genetic Algorithm (GA)
 - ❑ Particle Swarm Optimization (PSO)
 - ❑ Ant Colony Optimization (ACO)
 - ❑ Artificial Bee Colony Optimization (ABCO)
 - ❑ Cuckoo Search (CS)
 - ❑



Introduction To Genetic Algorithm (GA)

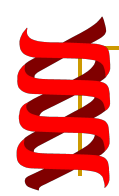
What Are Genetic Algorithms (GA)?

GA is a *search* and *optimization* technique based on Darwin's Principle of *Natural Selection*.



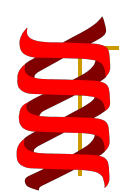
Darwin's Principle Of Natural Selection

- IF there are organisms that reproduce, and
- IF offsprings inherit traits (genetically determined characteristics) from their progenitors (ancestors or parents), and
- IF there is variability of traits, and
- IF the environment cannot support all members of a growing population,
- THEN those members of the population with less-adaptive traits (determined by the environment) will die out, and
- THEN those members with more-adaptive traits (determined by the environment) will thrive.



Basic Idea Of Principle Of Natural Selection

“Select The Best, Discard The Rest”



An Example of Natural Selection

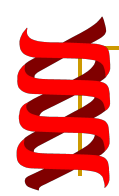
■ Giraffes have long necks.

Giraffes with slightly longer necks could feed on leaves of higher branches when all lower ones had been eaten off.

- ☐ They had a better chance of survival.
- ☐ Favorable behaviors propagated through generations of giraffes.
- ☐ Now, evolved species has long necks.

NOTE: Longer necks may have been a deviant characteristic (**mutation**) initially but since it was favorable, was propagated over generations.

So, some mutations are beneficial.



Evolution Through Natural Selection

Initial Population Of Animals



Struggle For Existence-Survival Of the Fittest



Surviving Individuals Reproduce, Propagate Favorable Characteristics

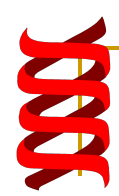


Evolved Species

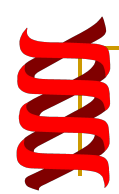
(Favorable Characteristic Now A Trait Of Species)

Millions Of Years

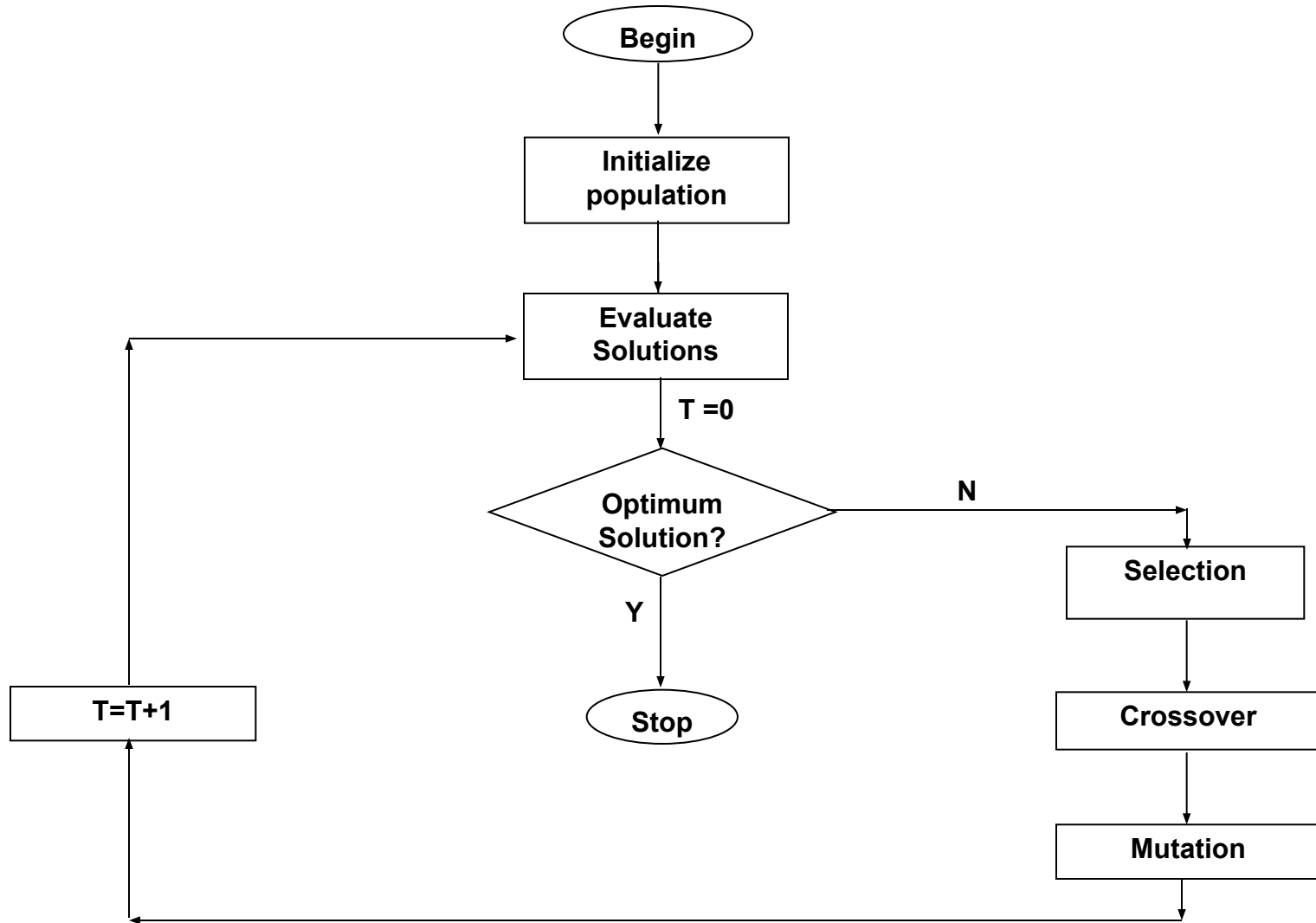




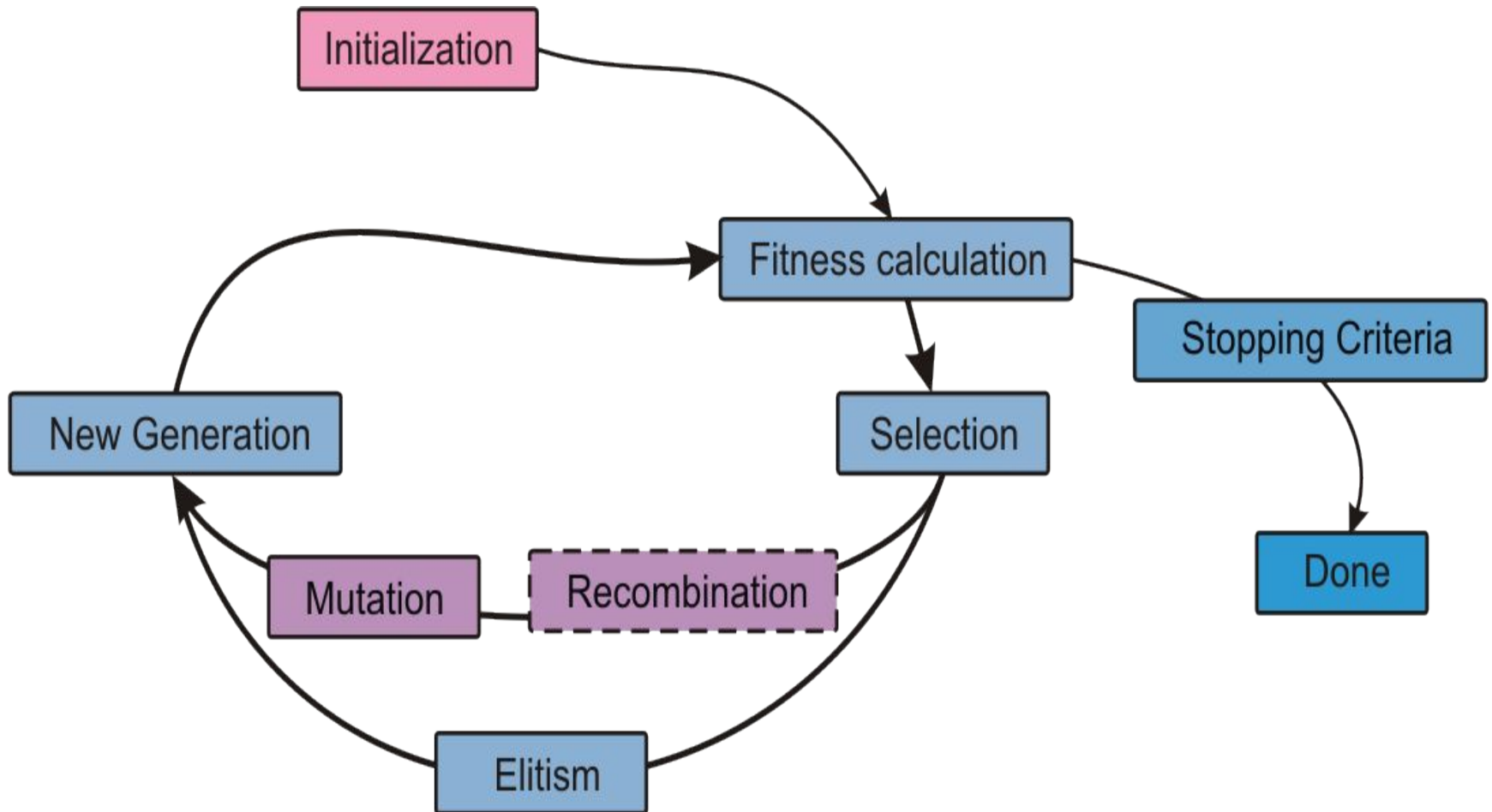
GA Implement Optimization Strategies By Simulating Evolution Of Species Through Natural Selection

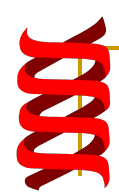


Working Mechanism Of GAs



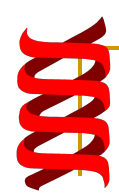
The Cycle of Evolution





Simple Genetic Algorithm

```
Simple_Genetic_Algorithm()  
{  
    Initialize the Population;  
    Calculate Fitness Function;  
  
    While(Fitness Value != Optimal Value)  
    {  
        Selection;//Natural Selection, Survival Of the  
Fittest  
        Crossover;//Reproduction, Propagate favorable  
characteristics  
  
        Mutation;//Mutation  
    }  
    Calculate Fitness Function;  
}
```

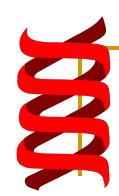


Nature to Computer Mapping

Nature	Computer
Population	Set of solutions.
Individual	Solution to a problem.
Fitness	Quality of a solution.
Chromosome	Encoding for a solution.
Gene	Part of the encoding of a solution.
Reproduction	Crossover



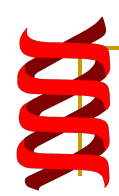
GA Operators and Parameters



Encoding

*The process of representing the solution in the form of a **string** that conveys the necessary information.*

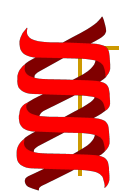
- Just as in a chromosome, each gene controls a particular characteristic of the individual, similarly, each bit in the string represents a characteristic of the solution.



Encoding Methods

- **Binary Encoding** – Most common method of encoding. Chromosomes are strings of 1s and 0s and each position in the chromosome represents a particular characteristic of the problem.

Chromosome A	10110010110011100101
Chromosome B	11111110000000011111



Encoding Methods (contd.)

- **Permutation Encoding** – Useful in ordering problems such as the Traveling Salesman Problem (TSP).
- **Example.** In TSP, every chromosome is a string of numbers, each of which represents a city to be visited.

Chromosome A	1 5 3 2 6 4 7 9 8
Chromosome B	8 5 6 7 2 3 1 4 9



Encoding Methods (contd.)

- **Value Encoding** – Used in problems where complicated values, such as real numbers, are used and where binary encoding would not suffice.

Good for some problems, but *often necessary to develop some specific crossover and mutation techniques for these chromosomes.*

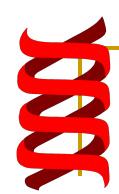
Chromosome A	1.235 5.323 0.454 2.321 2.454
Chromosome B	(left), (back), (left), (right), (forward)



Fitness Function

A fitness function quantifies the optimality of a solution (chromosome) so that particular solution may be ranked against all the other solutions.

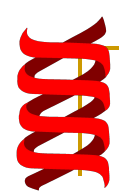
- A fitness value is assigned to each solution depending on how close it actually is to solving the problem.
- An ideal fitness function should correlate closely with the problem's goal, accurately measuring how well a solution solves the objective, and be quickly computable.
- **Example.** In TSP, $f(x)$ is sum of distances between the cities in solution. The lesser the value, the fitter the solution is.



Selection

The process that determines which solutions are to be preserved and allowed to reproduce and which ones deserve to die out.

- The primary objective is to **emphasize the good solutions** and **eliminate the bad solutions** in a population, **while keeping the population size constant**.
- “Selects The Best, Discards The Rest”.



Selection

- Identify the good solutions in a population.
- Make multiple solutions from the good solutions.
- Eliminate bad solutions from the population so that multiple copies of good solutions can be placed in the population.



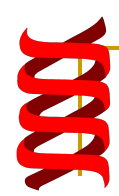
Roulette Wheel Selection

- Each current string in the population has a slot assigned to it which is in **proportion to its fitness**.
- We spin the weighted *roulette wheel* thus defined n times (where n is the total number of solutions).
- Each time Roulette Wheel stops, the string corresponding to that slot is created.

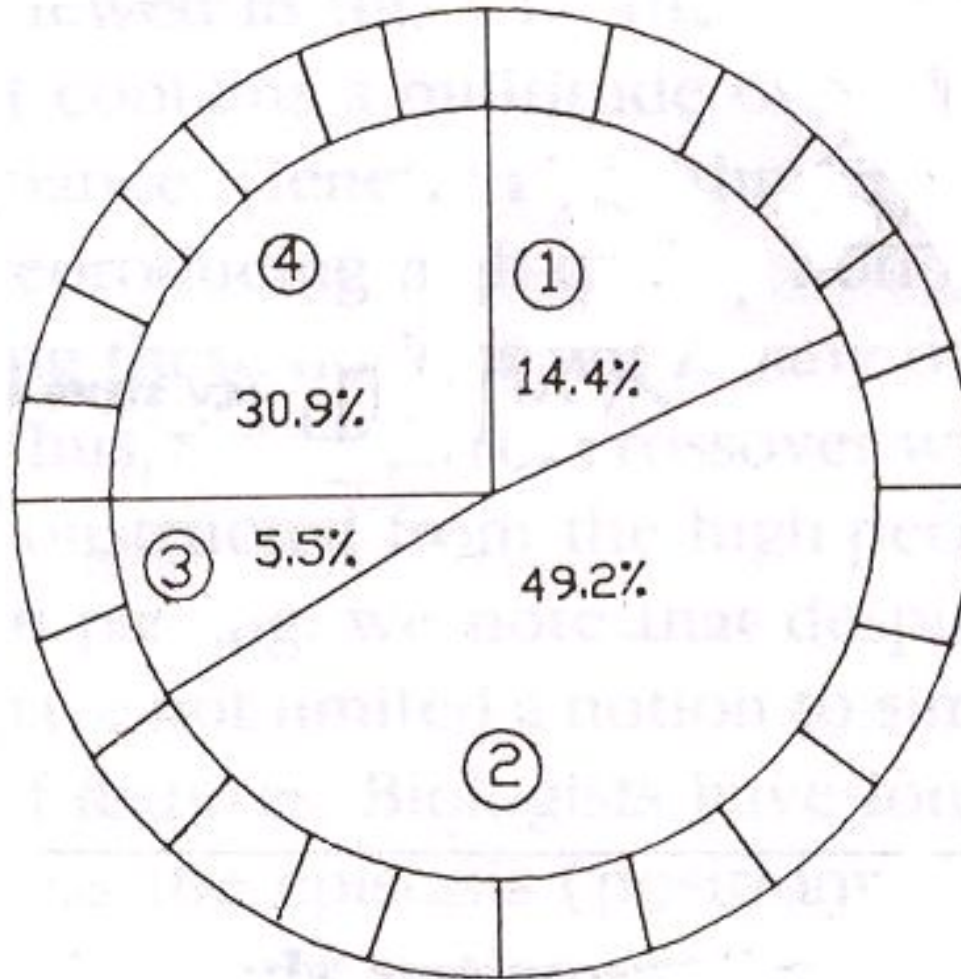
Strings that are fitter are assigned a larger slot and hence have a better chance of appearing in the new population.

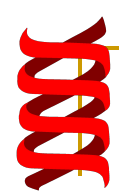
Example Of Roulette Wheel Selection

No.	String	Fitness	% Of Total
1	01101	169	14.4
2	11000	576	49.2
3	01000	64	5.5
4	10011	361	30.9
Total		1170	100.0



Roulette Wheel For Example

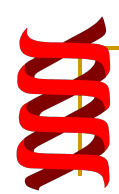




Crossover

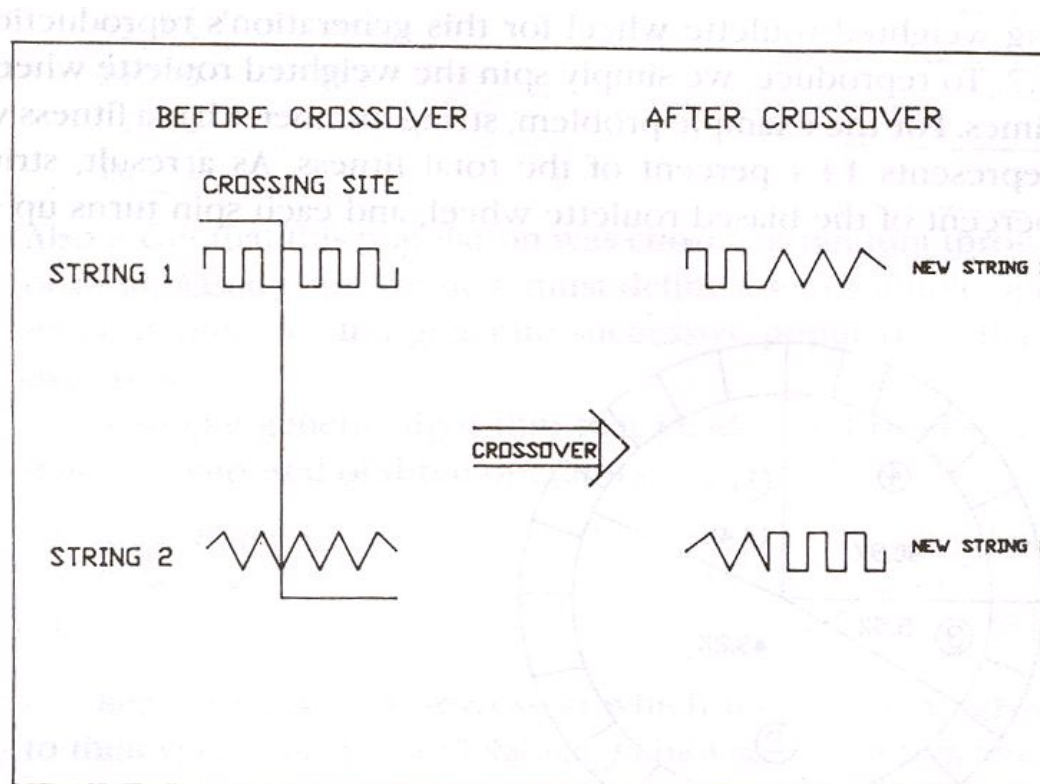
It is the process in which two chromosomes (strings) combine their genetic material (bits) to produce a new offspring which possesses both their characteristics.

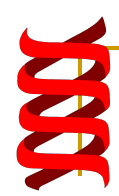
- *The method chosen depends on the Encoding Method.*



Crossover Methods

- **Single Point Crossover-** A random point is chosen on the individual chromosomes (strings) and the genetic material is exchanged at this point.

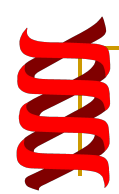




Crossover Methods (contd.)

■ Single Point Crossover

Chromosome1	11011 00100110110
Chromosome 2	11011 11000011110
Offspring 1	11011 11000011110
Offspring 2	11011 00100110110

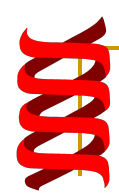


Crossover Methods (contd.)

- **Two-Point Crossover-** Two random points are chosen on the individual chromosomes (strings) and the genetic material is exchanged at these points.

Chromosome1	11011 00100 110110
Chromosome 2	10101 11000 011110
Offspring 1	10101 00100 011110
Offspring 2	11011 11000 110110

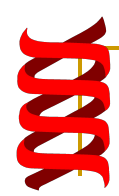
NOTE: These chromosomes are different from the last example.



Crossover Methods (contd.)

- **Uniform Crossover-** Each gene (bit) is selected randomly from one of the corresponding genes of the parent chromosomes.

Chromosome1	11011 00100 110110
Chromosome 2	10101 11000 011110
Offspring	10111 00000 110110



Crossover (contd.)

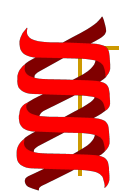
- Crossover between 2 good solutions **MAY NOT ALWAYS** yield a better or as good a solution.
- Since parents are good, probability of the child being good is high.
- If offspring is not good (poor solution), it will be removed in the next iteration during “Selection”.
- *Crossover Probability*: determines the number of chromosomes will take part in crossover.



Elitism

Elitism is a method which copies the best chromosome to the new offspring population before crossover and mutation.

- When creating a new population by crossover or mutation the best chromosome might be lost.
- Forces GAs to retain some number of the best individuals at each generation.
- Has been found that elitism significantly improves performance.

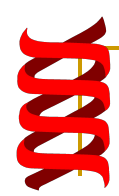


Mutation

It is the process by which a string is deliberately changed so as to maintain diversity in the population set.

We saw in the giraffes' example, that mutations could be beneficial.

Mutation Probability- determines how often the parts of a chromosome will be mutated.



Example Of Mutation

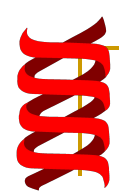
- For chromosomes using Binary Encoding, randomly selected bits are inverted.

Offspring	1101 1 00100 1 1 0110
Mutated Offspring	1101 0 00100 1 0 0110

NOTE: The number of bits to be inverted depends on the Mutation Probability.

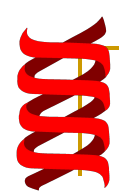
When to Use a GA

- Alternate solutions are too slow or overly complicated
- Problem is similar to one that has already been successfully solved by using a GA
- Want to hybridize with an existing solution



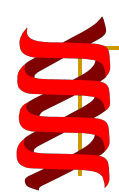
Simple Example

- **Simple problem: $\max x^2$ over $\{0,1,\dots,31\}$**
- GA approach:
 - Representation: binary code, e.g. $01101 \leftrightarrow 13$
 - Population size: 4
 - 1-point xover, bitwise mutation
 - Roulette wheel selection
 - Random initialization
- We show one generational cycle done by hand



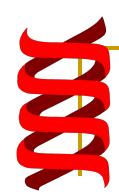
Selection

String no.	Initial population	x Value	Fitness $f(x) = x^2$	$Prob_i$	Expected count	Actual count
1	0 1 1 0 1	13	169	0.14	0.58	1
2	1 1 0 0 0	24	576	0.49	1.97	2
3	0 1 0 0 0	8	64	0.06	0.22	0
4	1 0 0 1 1	19	361	0.31	1.23	1
Sum			1170	1.00	4.00	4
Average			293	0.25	1.00	1
Max			576	0.49	1.97	2



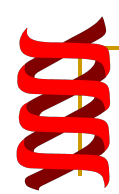
Crossover

String no.	Mating pool	Crossover point	Offspring after xover	x Value	Fitness $f(x) = x^2$
1	0 1 1 0 1	4	0 1 1 0 0	12	144
2	1 1 0 0 0	4	1 1 0 0 1	25	625
2	1 1 0 0 0	2	1 1 0 1 1	27	729
4	1 0 0 1 1	2	1 0 0 0 0	16	256
Sum					1754
Average					439
Max					729



Mutation

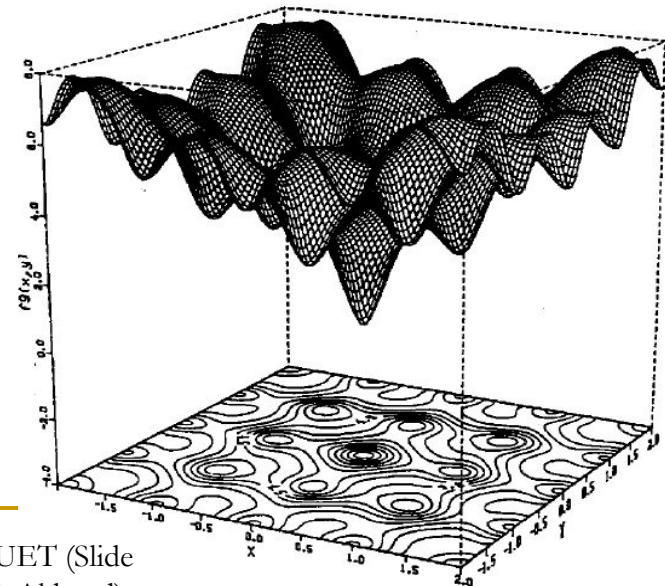
String no.	Offspring after xover	Offspring after mutation	x Value	Fitness $f(x) = x^2$
1	0 1 1 0 0	1 1 1 0 0	26	676
2	1 1 0 0 1	1 1 0 0 1	25	625
2	1 1 0 1 1	1 1 0 1 1	27	729
4	1 0 0 0 0	1 0 1 0 0	18	324
Sum				2354
Average				588.5
Max				729



Hard Problems

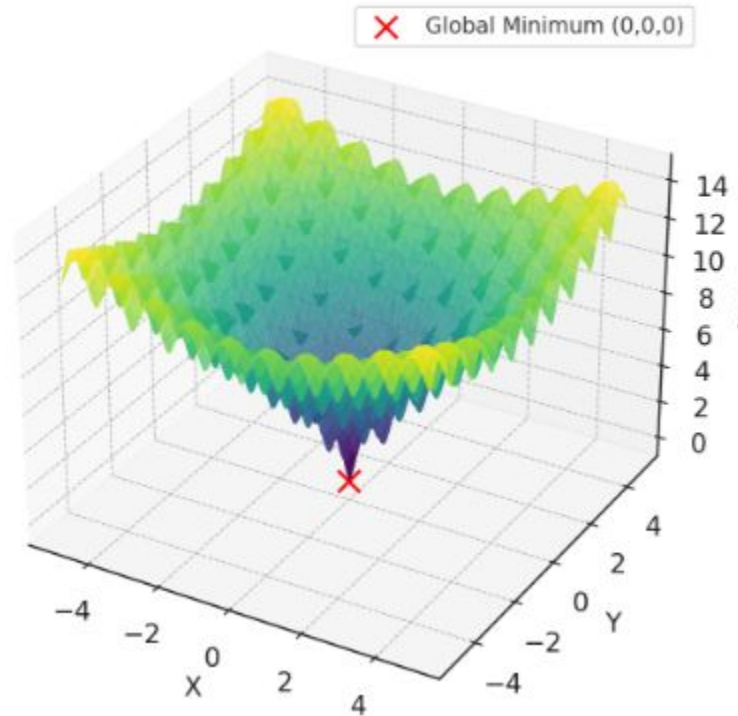
- Solution of $f(x)=x^2$ is easy to find and not realistic for GA
- Many problems occur as real valued problems, e.g., continuous parameter optimization $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- Illustration: Ackley's function (often used in EC)

$$f(\bar{x}) = -c_1 \cdot \exp \left(-c_2 \cdot \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \right) - \exp \left(\frac{1}{n} \cdot \sum_{i=1}^n \cos(c_3 \cdot x_i) \right) + c_1 + 1$$
$$c_1 = 20, c_2 = 0.2, c_3 = 2\pi$$



Hard Problems (Cont.)

Ackley's Function with Global Minimum





Genetic Algorithms To Solve The Traveling Salesman Problem (TSP)

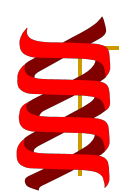


The Problem

The **Traveling Salesman Problem** is defined as:

‘We are given a set of cities and a symmetric distance matrix that indicates the cost of travel from each city to every other city.

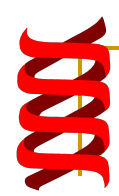
*The goal is to find **the shortest circular tour**, visiting every city exactly once, so as to **minimize the total travel cost**, which includes the cost of traveling from the last city back to the first city’.*



Encoding

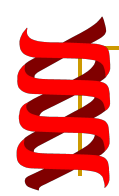
- Represent every city with an integer.
- Consider 6 Indian cities –
Mumbai, Nagpur, Calcutta, Delhi, Bangalore and Chennai and assign a number to each.

Mumbai	1 →
Nagpur	2 →
Calcutta	3 →
Delhi	4 →
Bangalore	5 →
Chennai	6 →



Encoding (contd.)

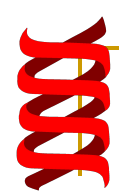
- Thus, a path would be represented as a **sequence** of integers from 1 to 6.
- The path **[1 2 3 4 5 6]** represents a path from Mumbai to Nagpur, Nagpur to Calcutta, Calcutta to Delhi, Delhi to Bangalore, Bangalore to Chennai, and finally from Chennai to Mumbai.
- This is an example of **Permutation Encoding** as the position of the elements determines the fitness of the solution.



Fitness Function

- The fitness function will be the **total cost of the tour** represented by each chromosome.
- This can be calculated as the **sum of the distances** traversed in each travel segment.

*The **Lesser The Sum, The Fitter The Solution** Represented By That Chromosome.*



Distance/Cost Matrix For TSP

	1	2	3	4	5	6
1	0	863	1987	1407	998	1369
2	863	0	1124	1012	1049	1083
3	1987	1124	0	1461	1881	1676
4	1407	1012	1461	0	2061	2095
5	998	1049	1881	2061	0	331
6	1369	1083	1676	2095	331	0

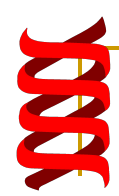
Cost matrix for six city example.

Distances in Kilometers



Fitness Function (contd.)

- So, for a chromosome [4 1 3 2 5 6], the total cost of travel or fitness will be calculated as shown below
- $$\text{Fitness} = 1407 + 1987 + 1124 + 1049 + 331 + 2095$$
$$= 7993 \text{ kms.}$$
- Since our objective is to **Minimize** the distance, the lesser the total distance, the fitter the solution.

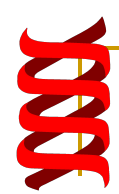


Selection Operator

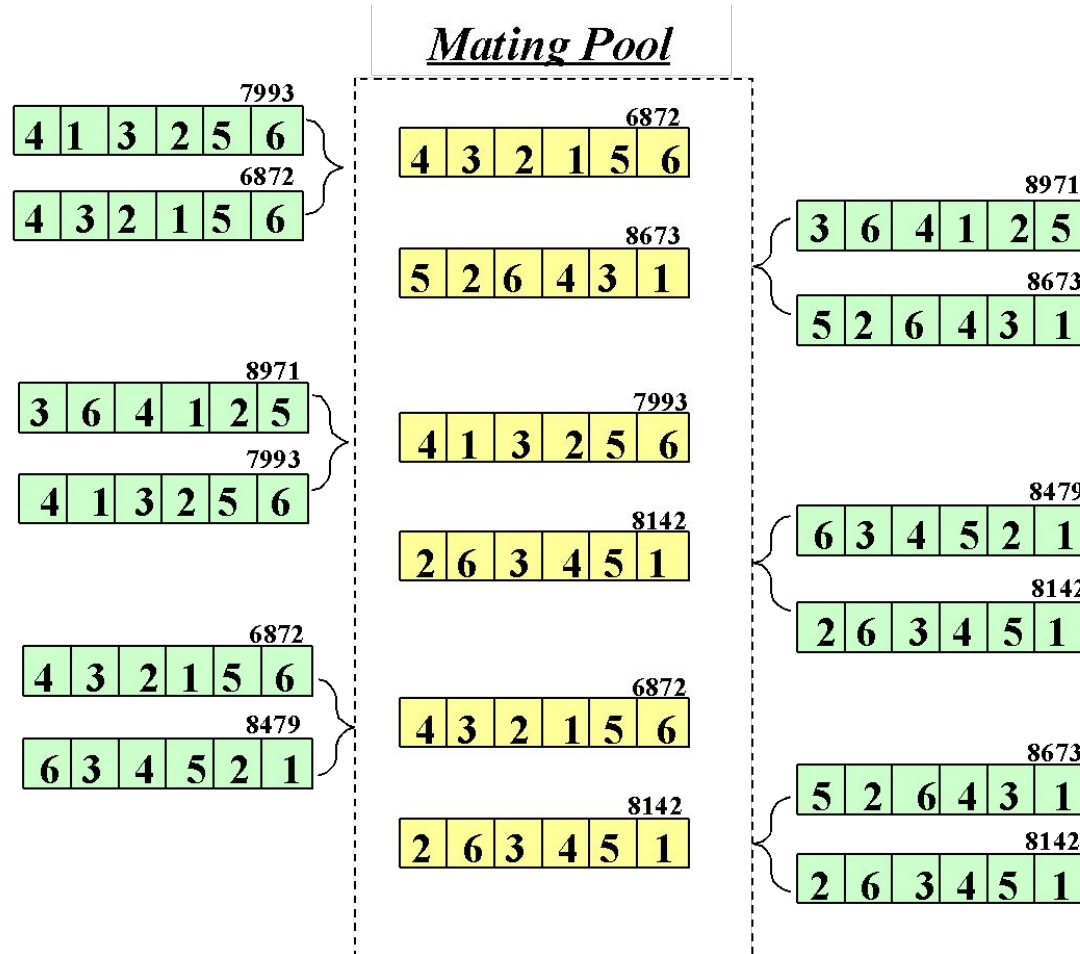
May use *Tournament Selection*.

As the name suggests *tournaments* are played between two solutions and the better solution is chosen and placed in the *mating pool*.

Two other solutions are picked again and another slot in the *mating pool* is filled up with the better solution.

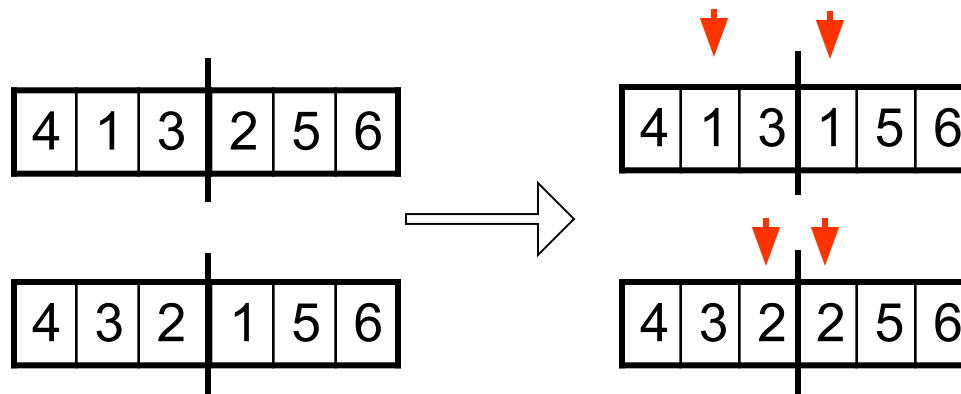


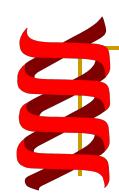
Tournament Selection (contd.)



Why we cannot use single-point crossover:

- Single point crossover method randomly selects a crossover point in the string and swaps the substrings.
- This may produce some **invalid offsprings** as shown below.





Crossover Operator

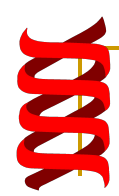
- Used the *Enhanced Edge Recombination* operator (T.Starkweather et al., 'A Comparison of Genetic Sequencing Operators', International Conference of GAs, 1991).
- This operator is different from other genetic sequencing operators in that it emphasizes *adjacency information* instead of the order or position of items in the sequence.
- The algorithm for the Edge-Recombination Operator involves constructing an *Edge Table* first.



Edge Table

The *Edge Table* is an *adjacency table* that lists links *into* and *out of* a city found in the two parent sequences.

If an item is already in the edge table and we are trying to insert it again, that element of a sequence must be a *common edge* and is represented by inverting its sign.



Finding The Edge Table

Parent 1

4	1	3	2	5	6
---	---	---	---	---	---

Parent 2

4	3	2	1	5	6
---	---	---	---	---	---

1	4	3	2	5
2	-3	5	1	
3	1	-2	4	
4	-6	1	3	
5	1	2	-6	
6	-5	-4		



Example Of Enhanced Edge Recombination Operator

Step 1

1	4	3	2	5
2	-3	5	1	
3	1	-2	4	
4	-6	1	3	
5	3	2	-6	
6	-5	-4		

↓

4					
---	--	--	--	--	--

Step 2

1	3	2	5
2	-3	5	1
3	1	-2	
4	-6	1	3
5	3	2	-6
6	-5		

↓

4	6				
---	---	--	--	--	--

Example Of Enhanced Edge Recombination Operator (contd.)

Step 3

1	3	2	5
2	-3	5	1
3	1	-2	
4	1	3	
5	3	2	
6	-5		

↓

4	6	5			
---	---	---	--	--	--

Step 4

1	3	2
2	-3	1
3	1	-2
4	1	3
5	3	2
6		

↓

4	6	5	3		
---	---	---	---	--	--

Example Of Enhanced Edge Recombination Operator (contd.)

Step 5

1	2	
2	1	
3	1	-2
4	1	
5	2	
6		

↓

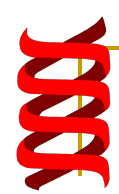
4	6	5	3	2	
---	---	---	---	---	--

Step 6

1	
2	1
3	1
4	1
5	
6	

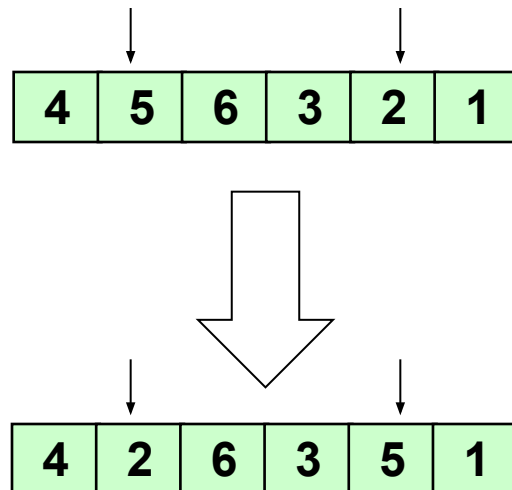
↓

4	6	5	3	2	1
---	---	---	---	---	---



Mutation Operator

- The mutation operator induces a change in the solution, so as to maintain diversity in the population and prevent *Premature Convergence*.
- In our project, we mutate the string by randomly selecting any two cities and interchanging their positions in the solution, thus giving rise to a new tour.



GA Applications

Domain	Application Types
Control	gas pipeline, pole balancing, missile evasion, pursuit
Design	semiconductor layout, aircraft design, keyboard configuration, communication networks
Scheduling	manufacturing, facility scheduling, resource allocation
Robotics	trajectory planning
Machine Learning	designing neural networks, improving classification algorithms, classifier systems
Signal Processing	filter design
Game Playing	poker, checkers, prisoner's dilemma
Combinatorial Optimization	set covering, travelling salesman, routing, bin packing, graph colouring and partitioning